

Antibiotic exposure in neonates and early adverse outcomes: a systematic review and meta-analysis

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Objectives: To systematically review and meta-analyse the relationship between antibiotic exposure in neonates and the following early adverse outcomes: necrotizing enterocolitis (NEC), invasive fungal infections (IFIs) and/or death.

Methods: Data sources were PubMed, Embase, Medline and the Cochrane Database (to December 2016), supplemented by manual searches of reference lists. Randomized controlled trials (RCTs) and observational studies were included if they provided data on different categories of antibiotic exposures (yes versus no, long versus short duration, and/or broad- versus narrow-spectrum regimens) and the risk of developing NEC, IFI and/or death in the neonatal period. Two reviewers extracted data and evaluated the risk of bias using the Cochrane Handbook, adapted to include observational studies. When appropriate, meta-analyses were conducted using the random-effect model.

Results: We identified 9 RCTs and 38 observational studies. The quality of the majority of studies was poor to moderate. There was a significant association between prolonged antibiotic exposure and an increased risk of NEC in five observational studies (5003 participants) and/or risk of death in five observational studies (13 534 participants). Eleven of 15 studies with data on broad- versus narrow-spectrum regimens reported an increased risk of IFI after broad-spectrum antibiotic exposure, in particular with third-generation cephalosporins and carbapenems. Meta-analysis was limited by few and old RCTs, insufficient sample sizes and diversity of antibiotic exposure and outcomes reported.

Conclusions: Prolonged antibiotic exposure in uninfected preterm infants is associated with an increased risk of NEC and/or death, and broad-spectrum antibiotic exposure is associated with an increased risk of IFI.

Introduction

Since the introduction of antibiotics in neonatal medicine, mortality from neonatal sepsis has declined.¹ However, infections are still an important cause of morbidity and mortality in term infants. Moreover, in preterm infants late-onset sepsis (LOS) is the main cause of death after 2 weeks of life,² and only a small reduction in infection rates has been observed in the last few decades.³ Clinical suspicion and fear of the potential dramatic consequences of neonatal sepsis if diagnosis is delayed result in the empirical use of antibiotics in many uninfected neonates.⁴ Antibiotics are the most commonly prescribed medications in the neonatal unit,⁵ and

virtually all extremely low birth-weight (ELBW; <1000 g) infants receive antibiotics during their first postnatal days, although the incidence of culture-proven early-onset sepsis (EOS) is extremely low in this population.^{6,7}

Clinicians tend to overuse antibiotics in neonates despite the existence of several guidelines on the appropriate use of antibiotics.^{8–10} Unnecessary antibiotic treatment in the neonatal period disturbs the microbial flora colonizing the neonate, and may lead to colonization with MDR bacteria.¹¹ Furthermore, antibiotic exposure early in life has been associated with adverse outcomes in the

neonatal period, but also with allergic diseases, obesity and inflammatory bowel disease in later life.^{12–14}

The purpose of the current systematic review is to identify, critically appraise and synthesize evidence from studies reporting different categories of antibiotic exposure in neonates and the subsequent risk of developing the following three adverse outcomes in the neonatal period: necrotizing enterocolitis (NEC), invasive fungal infections (IFIs) and/or death. We have included both observational studies and randomized controlled trials (RCTs), in line with suggestions from the Cochrane group stating that systematic reviews of rare adverse effects will usually need to include non-randomized studies in addition to RCTs, as the latter often primarily focus on effectiveness and not adverse effects.^{15,16}

Methods

This review is reported according to the preferred reporting items for systematic reviews and meta-analyses (PRISMA)¹⁷ following a registered protocol (study protocol registration: PROSPERO CRD42015026743).¹⁸ Our primary research question was 'Are different types of antibiotic exposure in neonates associated with increased risks of the adverse outcomes NEC, IFI and/or death in the neonatal period?'

Study selection and eligibility criteria

A study was eligible for review if it reported on groups of neonates, preterm and/or term, with different categories of intravenous antibiotic exposure and the adverse clinical outcomes NEC, IFI and/or death occurring in the neonatal period or up to discharge from the neonatal unit. We considered both non-randomized observational studies (cohort, case-control, cross-sectional) and RCTs. We compared three different categories of antibiotic exposure: (i) yes versus no antibiotics; (ii) long versus short duration of antibiotic therapy; and (iii) broad-spectrum versus narrow-spectrum antibiotic regimens. For category (ii), we suggested in advance that 'prolonged' antibiotic exposure was always ≥ 3 days or the longest regimen among two antibiotic regimens compared. For category (iii), we always defined regimens including third-generation cephalosporins or carbapenems as a broad-spectrum regimen when compared with regimens containing aminoglycosides for coverage against Gram-negative bacteria. This definition was also based on the fact that empirical treatment using a third-generation cephalosporin for Gram-negative coverage induces significantly more antibiotic resistance than a regimen containing an aminoglycoside.¹¹ NEC was defined as Bell's stage 2–3.¹⁹ IFI was defined as fungaemia or detection of fungi in otherwise sterile body sites. Death, as an adverse outcome, was defined as any cause of death, including death attributed to infection during antibiotic therapy in the neonatal period or up to discharge from the neonatal unit. If infants were born prematurely we defined the neonatal period as up to 44 weeks post-menstrual age. We included case-control studies reporting on prespecified adverse outcomes if data on antibiotic exposure prior to the outcomes were presented as extractable data in cases and controls, respectively. We excluded studies investigating antenatal antibiotics, oral antibiotics, low-dose intravenous vancomycin prophylaxis in preterm neonates and studies with a non-neonatal population.

Search strategy

Our search strategy was developed in consultation with an epidemiologist, a librarian, a paediatric pharmacologist and a neonatologist (Appendix A, available as Supplementary data at JAC Online (<http://jac.oxfordjournals.org/>)). We searched PubMed, Embase, Medline and the Cochrane Database using MeSH terms and text words from the inception of each database up to December 2016. The target was human studies written in English. There

was no publication period restriction. We did not contact authors for supplementary information and we did not perform searches in the grey literature. The first search was conducted using MeSH terms. The search strategy in PubMed, Medline and the Cochrane Database was as follows: 'Infant, Newborn' and 'Anti-Bacterial Agents' with one of the following outcome terms: 'Enterocolitis, Necrotizing', 'Fungemia', 'Candidiasis, Invasive', 'Meningitis, Fungal' or 'Mortality'. The Embase database uses its own keywords, and 'Newborn' and 'Antibiotic Agent' were combined with one of the following outcome terms: 'Necrotising Enterocolitis', 'Fungemia', 'Invasive Candidiasis', 'Fungal Meningitis' or 'Mortality'. The second search was conducted using free text in PubMed, Medline and Embase combining the keywords 'Infant, Low Birth Weight', 'Infant, Postmature', 'Infant, Premature' or 'Infant, Newborn' with 'Anti-Bacterial Agents' or 'Antibiotics', and one of the following: 'Necrotizing Enterocolitis' or 'Fungaemia' or 'Fungemias' or 'Candidemia' or 'Invasive Candidiasis' or 'Fungal Meningitis' or 'Mortality'. Finally, we looked at reference lists and citations of included studies and relevant previous reviews to identify any additional eligible studies. All citations were then combined and duplicates/triplicates were excluded.

Screening, data extraction and management

Search results were independently screened in duplicate, and two reviewers (E. E. and J. W. F.) assessed each potentially eligible full-text article according to our predetermined inclusion and exclusion criteria. Only studies mutually agreed as irrelevant were excluded. A third reviewer (C. K.) had the decisive vote in case of disagreement. The following information was extracted from the articles: author, year and country; study design; study population, including gestational age (GA) and birth weight (BW), comparison of outcomes between groups with different categories of antibiotic exposure, and, if available, risk estimates with 95% CI for the specific outcome.

Assessment of methodological quality

Methodological quality was assessed by using the Cochrane Handbook of Systematic Reviews of Interventions¹⁵ and recently published suggestions on how to assess risk of bias and confounding in observational studies.²⁰ Five domains related to risk of bias were assessed for each study included: selection, performance, detection, reporting and confounding. Two reviewers (E. E. and C. K.) assessed the methodological quality of each study. Risks of bias were judged as low, high or unclear (Supplementary data Appendix B). Disagreements in the categorization process were resolved after discussion.

Synthesis of results and analysis

We classified studies according to their adverse outcomes, including the comparisons of different categories of antibiotic exposure. Most non-randomized studies are reported separately and were not pooled for meta-analysis because of marked clinical and methodological diversity regarding e.g. interventions, antibiotics used, study design and the outcomes reported. We combined adverse outcomes of interest from studies considered sufficiently homogeneous to provide a meaningful summary, and calculated combined effect estimates. In the meta-analyses we pooled RCTs and non-randomized studies, the latter only if clinical baseline characteristics of patient groups that experienced different antibiotic exposures (categories 1–3) were similar and the studies reported dichotomous outcomes. Subgroup analysis was performed for RCTs and non-randomized studies. We quantified inconsistency between the results of the studies by using the I^2 test. Interpretation of thresholds for statistical heterogeneity was as follows: I^2 values between 0% and 40% might not be important, whereas higher I^2 values may represent moderate (30%–60%), substantial (50%–90%) or considerable heterogeneity (75%–100%).¹⁵ Data entry and meta-analysis were performed using RevMan version 5.3 (The Nordic

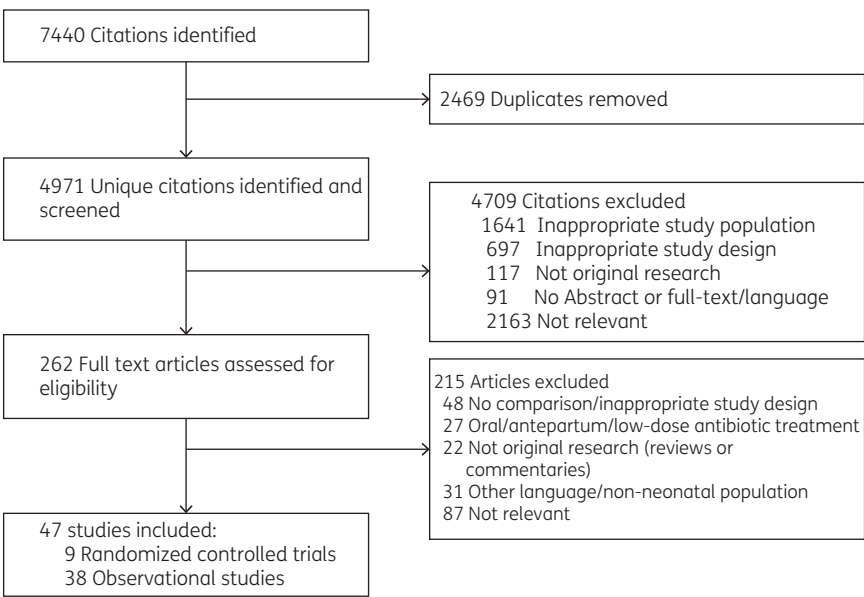


Figure 1. Preferred reporting items for systematic reviews and meta-analysis (PRISMA) flow diagram.

Cochrane Centre, Copenhagen, Denmark). We calculated ORs with 95% CI for the outcomes of interest. We present the effect estimates by using the random-effect model due to assumption of clinical and methodological diversity among the studies, subsequently often leading to statistical heterogeneity.

Results

Overview of included studies

From 4972 unique citations identified, 262 potentially eligible full-text articles were reviewed and 47 studies met our inclusion criteria: 9 RCTs^{21–29} published between 1983 and 2014, and 38 observational non-randomized studies^{4,7,30–65} published between 1980 and 2016 (Figure 1). Among the 38 non-randomized studies there were 17 case-control studies, 18 retrospective cohort studies and three prospective cohort studies. There was a large diversity between the studies regarding antibiotics used, as well as onset and duration of antibiotic exposure after birth. Tables 1–3 list the main characteristics and the primary adverse outcomes of interest retrieved from all 47 included studies. Tables S1–S3 (available as Supplementary data at JAC Online) give detailed descriptions of all 47 studies, including outcomes.

Risk of bias of included studies

Diversity in study design and reporting made quality evaluation challenging, particularly for non-randomized studies. We defined risk of reporting bias as unclear or high if data concerning one or two of the pre-specified outcomes were not reported. The majority of RCTs and non-randomized studies were of moderate to poor quality (Figure 2a–c). Despite overall moderate to low scores, adjusted analyses, in order to counter potential selection bias, were reported in 24/38 non-randomized studies.^{7,30,32–35,38,40–44,46–48,50,51,56,57,60–64} Only one of the nine RCTs was registered in a public trial registry.²⁷ Blinding was not

performed or not reported in eight out of the nine RCTs. In one RCT the researchers evaluating the outcome were blinded.²⁵

Necrotizing enterocolitis

Twenty studies reported different categories of previous antibiotic exposure and subsequent risk of NEC (Table 1). Two studies reported outcome data concerning both duration of antibiotic exposure and antibiotic treatment—yes/no,^{30,54} the other 18 studies reported only outcome data from one category of antibiotic exposure. All RCTs had treatment success/failure of the antibiotic regimen as primary outcome, and NEC was only reported as a secondary adverse outcome. In 13 of the included studies, NEC was clearly defined using modifications of Bell’s criteria. Five studies reported composite outcomes of NEC or death^{33–35} or NEC, LOS or death.^{30,57}

Antibiotics: yes/no in NEC

Six studies, two small RCTs^{21,25} and four observational studies,^{32,54–56} were included. In the pooled analysis of the two RCTs (Figure 3a) the risk of NEC after antibiotic exposure was not significantly increased, but there was substantial heterogeneity between study results. Three case-control studies reported divergent results, with two studies^{53,54} presenting non-significantly more use of antibiotics in NEC cases than controls, whereas use of antibiotics in the first 2 days of life reduced the risk of NEC in one study.⁵²

Antibiotics: duration in NEC

Ten observational studies were included.^{30,33–35,54–57,62,65} A large retrospective cohort study by Cotten *et al.*³³ included 4039 ELBW infants. When adjusting for maternal, perinatal and neonatal variables previously shown to be associated with NEC and death, they found that prolonged empirical antibiotic therapy (≥4 days) after birth, in patients without proven EOS, was associated with

Table 1. Studies reporting previous antibiotic exposures and risk of NEC: summary of main characteristics and results

Source	Design	No. of participants	GA and BW	Antibiotic exposure and risk of NEC		
				no vs yes	short vs prolonged	narrow vs broad spectrum
Cantey <i>et al.</i> , ⁶⁵ 2016 (USA)	retrospective cohort	2502	all GAs	NDA	no difference	NDA
Greenwood <i>et al.</i> , ⁵⁷ 2014 (USA)	prospective cohort	74	GA ≤ 32 weeks	NDA	no difference	NDA
Chang <i>et al.</i> , ⁶² 2013 (South Korea)	retrospective cohort	99	BW < 1.5 kg	NDA	no difference	broader spectrum: ↑ risk of NEC
Chong <i>et al.</i> , ³⁶ 2013 (USA)	retrospective matched cohort	484	BW 0.5–1.5 kg	NDA	NDA	broader spectrum: ↓ risk of NEC
Shah <i>et al.</i> , ³⁵ 2013 (Australia)	retrospective cohort	216	GA < 28 weeks, survival > 3 days	NDA	no difference	NDA
Abdel Ghany and Ali, ³⁴ 2012 (Egypt)	retrospective cohort	207	BW < 1.5 kg, survival > 5 days	NDA	prolonged use: ↑ risk of NEC and/or death	NDA
Alexander <i>et al.</i> , ⁵⁶ 2011 (USA)	case-control	372	preterm, mean GA 28 weeks	NDA	prolonged use: ↑ risk of NEC	NDA
Kuppala <i>et al.</i> , ³⁰ 2011 (USA)	retrospective cohort	365	GA ≤ 32 weeks, BW ≤ 1.5 kg	no difference	prolonged use: no difference NEC (alone), but ↑ risk of NEC, LOS or death	NDA
Metsvaht <i>et al.</i> , ²⁷ 2010 (Estonia)	RCT	283	all GAs	NDA	NDA	no difference
Tagare <i>et al.</i> , ²⁵ 2010 (India)	RCT	140	preterm, GA < 37 weeks	no difference	NDA	NDA
Cotten <i>et al.</i> , ³³ 2009 (USA)	retrospective cohort	4039	BW ≤ 1 kg, survival > 5 days	NDA	prolonged use: ↑ risk of NEC and/or death	NDA
Wang <i>et al.</i> , ⁵⁵ 2009 (USA)	case-control	20	GA 25–32 weeks	NDA	prolonged use: ↑ risk of NEC	NDA
Clark <i>et al.</i> , ³⁷ 2006 (USA)	retrospective cohort	128914	all GAs, median GA 35 weeks	NDA	NDA	broader spectrum: ↓ risk of NEC
Allen and Da Silva, ⁵⁸ 2003 (Canada)	retrospective cohort	62	BW < 1 kg, survival > 4 days	NDA	NDA	no difference
Krediet <i>et al.</i> , ⁵² 2003 (Netherlands)	case-control	208	all GAs, median GA 29 weeks	early use: ↓ risk NEC	NDA	NDA
Harms <i>et al.</i> , ²¹ 1995 (Germany)	RCT	148	preterm, mean GA 29 weeks	no difference	NDA	NDA
Millar <i>et al.</i> , ²⁸ 1992 (England)	RCT	81	GA < 33 weeks	NDA	NDA	no difference
Mufti and Bhutta, ⁵³ 1992 (Pakistan)	case-control	39	BW ≤ 2 kg	no difference	NDA	NDA
Hall <i>et al.</i> , ²⁹ 1988 (England)	RCT	222	all GAs	NDA	NDA	no difference
Stoll <i>et al.</i> , ⁵⁴ 1980 (USA)	case-control	133	all GAs	no difference	no difference	NDA

NDA, no data available.

Table 2. Studies reporting previous antibiotic exposures and risk of IFI: summary of main characteristics and results

Source	Design	No. of participants	GA and BW	Antibiotic exposure and risk of IFI		
				no vs yes	short vs prolonged	narrow vs broader spectrum
Fu <i>et al.</i> , ⁶⁴ 2016 (China)	case-control	96	BW < 1.5 kg	NDA	no difference	broader spectrum: ↑ risk of IFI
Tewari and Jain, ²⁴ 2014 (India)	RCT	187	GA ≥ 28 weeks, BW ≥ 1 kg	NDA	NDA	no difference
Aliaga <i>et al.</i> , ⁶³ 2013 (USA)	retrospective cohort	709325	all GAS	NDA	NDA	broader spectrum: ↑ risk of IFI
Chang <i>et al.</i> , ⁶² 2013 (South Korea)	retrospective cohort	99	preterm, BW < 1.5 kg	NDA	no difference	broader spectrum: ↑ risk of IFI
Lee <i>et al.</i> , ⁵⁰ 2013 (USA)	retrospective cohort	530162	BW > 1.5 kg	NDA	NDA	broader spectrum: ↑ risk of IFI
Yu <i>et al.</i> , ⁴⁷ 2013 (China)	case-control	135	all GAS	NDA	no difference	broader spectrum: ↑ risk of IFI
Ariff <i>et al.</i> , ⁴⁰ 2011 (Pakistan)	case-control	81	all GAS	NDA	no difference	broader spectrum: ↑ risk of IFI
Benjamin <i>et al.</i> , ⁶¹ 2010 (USA)	retrospective cohort	1515	BW ≤ 1 kg, survival > 3 days	NDA	NDA	broader spectrum: ↑ risk of IFI
Benjamin <i>et al.</i> , ⁴⁹ 2006 (USA)	retrospective cohort	4579	BW ≤ 1 kg, survival > 3 days	NDA	NDA	broader spectrum: ↑ risk of IFI
Cotten <i>et al.</i> , ³² 2006 (USA)	retrospective cohort	3702	BW ≤ 1 kg, survival > 3 days	NDA	prolonged use: ↑ risk of IFI	broader spectrum: ↑ risk of IFI
Manzoni <i>et al.</i> , ⁴² 2006 (Italy)	nested case-control	201	preterm, BW < 1.5 kg	NDA	no difference	no difference
Feja <i>et al.</i> , ³⁸ 2005 (USA)	case-control	180	preterm, mean GA 30 weeks	no difference	NDA	NDA
Linder <i>et al.</i> , ⁵¹ 2004 (Israel)	case-control	112	preterm, mean GA 28–29 weeks	NDA	NDA	no difference
Auriti <i>et al.</i> , ²² 2005 (Italy)	RCT	130	GA < 32 weeks	NDA	no difference	NDA
Benjamin <i>et al.</i> , ⁴⁸ 2003 (USA)	retrospective cohort	6172	BW < 1.25 kg, survival ≥ 3 days	NDA	NDA	broader spectrum: ↑ risk of IFI
Pera <i>et al.</i> , ⁴³ 2002 (USA)	case-control	334	preterm, BW < 1.25 kg	NDA	prolonged use: ↑ risk of IFI	NDA
Warris <i>et al.</i> , ⁴⁵ 2001 (Netherlands)	case-control	24	GA ≤ 33 weeks	NDA	prolonged use: ↑ risk of IFI	NDA
Benjamin <i>et al.</i> , ⁶⁰ 2000 (USA)	case-control	51	preterm, mean GA 28 weeks and BW 1.1 kg	NDA	NDA	broader spectrum: ↑ risk of IFI
Saiman <i>et al.</i> , ⁴⁴ 2000 (USA)	prospective cohort	2847	all GAS, hospitalization ≥ 3 days	NDA	no difference	NDA
Singh <i>et al.</i> , ³⁹ 1999 (India)	prospective cohort	70	preterm	antibiotic use: ↑ risk of IFI	NDA	NDA
Lin <i>et al.</i> , ⁵⁹ 1998 (Taiwan)	case-control	358	BW < 1.5 kg, GA ≤ 33 weeks	NDA	no difference	NDA
Faix <i>et al.</i> , ⁴¹ 1989 (USA)	prospective cohort	41	BW ≤ 1.5 kg	NDA	prolonged use: ↑ risk of IFI	NDA
Weese-Mayer <i>et al.</i> , ⁴⁶ 1987 (USA)	case-control	55	all GAS, mean BW 1.9 kg and mean GA 32–33 weeks	NDA	prolonged use: ↑ risk of IFI	NDA
Snelling <i>et al.</i> , ²³ 1983 (England)	RCT	55	all GAS, mean BW 1.7 kg and mean GA 33 weeks	NDA	NDA	no difference

NDA, no data available.

Table 3. Studies reporting previous antibiotic exposures and risk of death: summary of main characteristics and results

Source	Design	No. of participants	GA and BW	Antibiotic exposure and risk of death		
				no vs yes	short vs prolonged	narrow vs broader spectrum
Canvey et al., ⁶⁵ 2016 (USA)	retrospective cohort	2502	all GAs	NDA	no difference	NDA
Fjalstad et al., ⁴ 2016 (Norway)	retrospective cohort	10175	GA ≥ 37 weeks	NDA	NDA	no difference
Ting et al., ⁷ 2016 (Canada)	retrospective cohort	8824	BW < 1.5 kg	NDA	prolonged use: ↑ risk of death	NDA
Greenwood et al., ⁵⁷ 2014 (USA)	prospective cohort	74	GA ≤ 32 weeks	NDA	no difference	NDA
Tewari and Jain, ²⁴ 2014 (India)	RCT	187	GA ≥ 28 weeks, BW ≥ 1 kg	NDA	NDA	no difference
Chang et al., ⁶² 2013 (South Korea)	retrospective cohort	99	BW < 1.5 kg	NDA	prolonged use: ↑ risk of death	broader spectrum: ↑ risk of death
Chong et al., ³⁶ 2013 (USA)	retrospective matched cohort	484	BW 0.5–1.5 kg	NDA	NDA	no difference
Shah et al., ³⁵ 2013 (Australia)	retrospective cohort	216	GA < 28 weeks, survival > 3 days	NDA	no difference	NDA
Abdel Ghany and Ali, ³⁴ 2012 (Egypt)	retrospective cohort	207	BW < 1.5 kg, survival > 5 days	NDA	prolonged use: ↑ risk of death	NDA
Kuppala et al., ³⁰ 2011 (USA)	retrospective cohort	365	GA ≤ 32 weeks, BW ≤ 1.5 kg	no difference	prolonged use: no difference death (alone), but ↑ risk of NEC, LOS or death	NDA
Metsvaht et al., ²⁷ 2010 (Estonia)	RCT	283	all GAs	NDA	NDA	no difference
Tagare et al., ²⁵ 2010 (India)	RCT	140	preterm, GA < 37 weeks	no difference	NDA	NDA
Cotten et al., ³³ 2009 (USA)	retrospective cohort	4039	BW ≤ 1 kg, survival > 5 days	NDA	prolonged use: ↑ risk of death	NDA
Clark et al., ³⁷ 2006 (USA)	retrospective cohort	128914	all GAs (median GA 29 weeks)	NDA	NDA	broader spectrum: ↑ risk of death
Cotten et al., ³² 2006 (USA)	retrospective cohort	3702	BW ≤ 1 kg, survival > 3 days	NDA	no difference	NDA
Allen and Da Silva, ⁵⁸ 2003 (Canada)	retrospective cohort	62	BW < 1 kg, survival > 4 days	NDA	NDA	no difference
Auriti et al., ²² 2005 (Italy)	RCT	130	GA < 32 weeks	NDA	no difference	NDA
Cordero et al., ³¹ 2003 (USA)	retrospective matched cohort	517	BW < 1 kg	NDA	no difference	NDA
Harms et al., ²¹ 1995 (Germany)	RCT	148	preterm, mean GA 29 weeks	no difference	NDA	NDA
De Louvois et al., ²⁶ 1992 (Europe)	RCT	1316	all GAs	NDA	NDA	no difference
Millar et al., ²⁸ 1992 (England)	RCT	81	GA < 33 weeks	NDA	NDA	no difference

NDA, no data available.

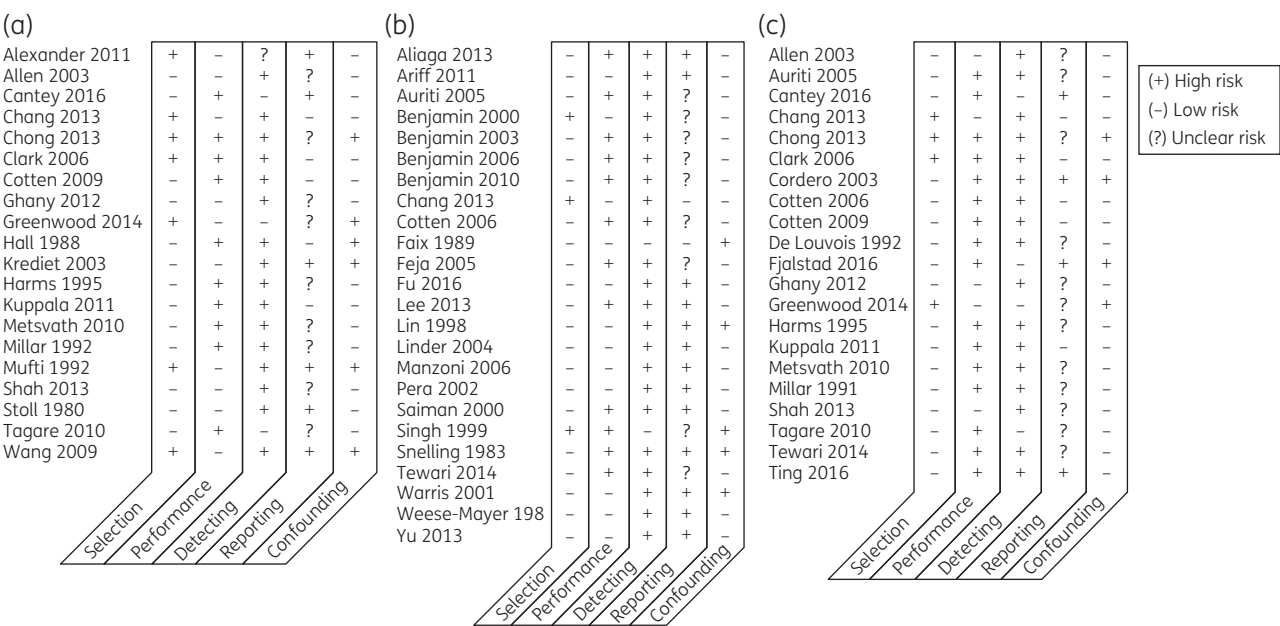


Figure 2. Risk of bias graph: review of authors’ judgements about each risk of bias item for each included study and the three outcomes. (a) Studies reporting on risk of NEC (n = 20). (b) Studies reporting on risk of IFIs (n = 24). (c) Studies reporting on risk of death (n = 21).

increased odds of NEC (OR 1.34, 95% CI 1.04–1.73) compared with shorter duration. These authors also investigated ≥ 5 days versus shorter duration of empirical antibiotic therapy and found increased odds of NEC or death (but not NEC alone) (OR 1.30, 95% CI 1.10–1.54) compared with shorter duration of antibiotic treatment. Four other studies,^{30,34,55,56} including 964 preterm infants, also showed associations between duration of antibiotic exposure and NEC, or the composite outcome NEC, LOS or death. In contrast, five studies did not show a significant difference in NEC rates. However, Cantey and co-workers⁶⁵ studied predominantly (81%) mature infants with GA >34 weeks, constituting a low-risk group for NEC. Three of the remaining four studies were small, but a trend towards higher rates of NEC in patients receiving prolonged antibiotic therapy was reported by Chang *et al.*⁶² (OR 7.69, 95% CI 0.93–63.3), Shah *et al.*³⁵ (OR 2.1, 95% CI 0.8–5.3) and Stoll *et al.*⁵⁴ (8.2 versus 6.8 days) (Table S1). Overall, prolonged antibiotic therapy was not defined consistently among the 10 studies in this category. Moreover, we did not find it appropriate to pool these studies as only 1 out of 10 studies had clearly defined comparable groups with different duration of antibiotic exposure.

Antibiotics: broad- versus narrow-spectrum regimen in NEC

Seven studies were included.^{27–29,36,37,58,62} A large retrospective cohort study by Clark *et al.*³⁷ reported on two patient groups receiving either ampicillin and cefotaxime or ampicillin and gentamicin. We did not find it appropriate to include this study in a meta-analysis due to clinical differences among patients in the two groups. However, there was a significantly higher rate of NEC (combined medical and surgical) in the group receiving ampicillin and gentamicin. The antibiotic regimens in the other six studies varied. Two studies^{27,58} reported no difference in the risk of NEC

comparing broad- versus more narrow-spectrum antibiotics. Chong *et al.*³⁶ compared ampicillin and gentamicin versus piperacillin/tazobactam for suspected EOS in very low birth-weight infants (VLBW; <1500 g) in a retrospective matched case-control study. In this before-after study, with a high risk of bias, the authors reported a much lower rate of NEC after implementing piperacillin/tazobactam as empirical therapy. We pooled six studies (three RCTs^{27–29} and three observational studies^{36,58,62}) with comparable groups. In the pooled analyses (Figure 3b) and in the subgroup analyses of RCTs and observational studies, there were no differences in risk of NEC with a broad- versus narrow-spectrum regimen. However, there was substantial heterogeneity between study results in both the pooled analysis and the subgroup analysis.

IFI

Twenty-four studies reported on the risk of IFI after antibiotic exposure in the neonatal period (Table 2). The majority assessed either antibiotic therapy duration or compared different antibiotic regimens. Only two small observational studies of moderate to poor quality assessed antibiotics—yes/no.^{40–41} Most studies (18 out of 24) focused on preterm infants. There was diversity in defining IFI between studies. Fourteen studies reported on candidaemia defined as a positive blood culture,^{22,24,38–40,43–46,48,51,59,60,64} whereas 7 studies reported on invasive candidiasis^{23,32,41,49,50,61,63} and 2 studies reported IFI without any further definitions.^{42,47} Invasive candidiasis was defined as having a positive culture with *Candida* species in blood or CSF in two studies, and as blood, urine, CSF or other sterile body fluids, such as peritoneal fluid in four studies, respectively. Manzoni *et al.*⁴² and Aliaga *et al.*⁶³ defined IFI as microbiologically proven culture in blood, urine or CSF.

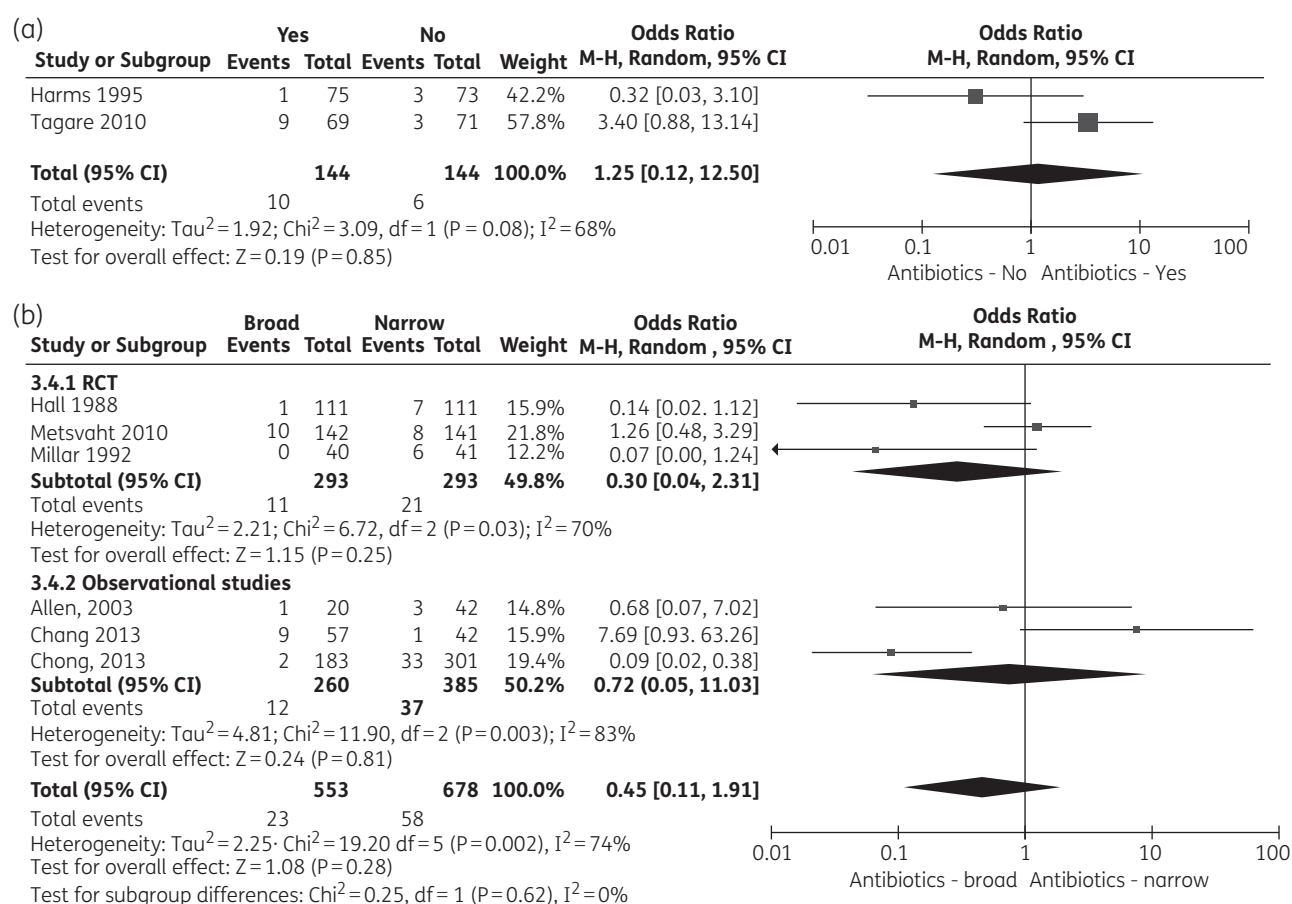


Figure 3. Forest plots stratified by antibiotic exposure and outcomes. (a) Pooled results of two studies comparing risk of necrotizing enterocolitis between children who received antibiotics or did not (yes versus no). (b) Pooled results of six studies comparing risk of NEC between children who received broad- versus narrow-spectrum antibiotic regimens. Subgroup analysis of RCTs and observational studies. (c) Pooled results of three studies comparing risk of death between children who received prolonged or shorter duration of antibiotic therapy. Subgroup analysis of RCTs and observational studies. (d) Pooled results of eight studies comparing risk of death between children who received broader- versus narrower-spectrum antibiotic regimens. Subgroup analysis of RCTs and observational studies. The sizes of the squares are proportional to study weights. Diamond markers indicate pooled effect sizes.

Antibiotics: duration in IFI

Thirteen studies were included.^{22,32,40–47,59,62,64} Five studies, published between 1987 and 2001, reported an increased risk of IFI after prolonged duration of antibiotic therapy (Table 2).^{32,41,43,45,46} In contrast, eight studies, published between 2000 and 2016, reported no increased risk of IFI after prolonged duration of antibiotic exposure.^{22,40,42,44,47,59,62,64}

Antibiotics: broad- versus narrow-spectrum regimen in IFI

Fifteen studies were included.^{23,24,32,40,42,47–51,60–64} Broad-spectrum antibiotics used in the majority of these studies were third-generation cephalosporins or carbapenems. Ten studies, published between 2000 and 2013, reported significantly increased risk of IFI after exposure to third-generation cephalosporins.^{32,40,47–50,60–63} Fu *et al.*⁶⁴ reported that previous use of carbapenems was significantly associated with an increased risk of candidaemia in a multivariate logistic regression analysis. Linder *et al.*⁵¹ reported broad-spectrum antibiotics, including

third-generation cephalosporins, as a predisposing factor for IFI in a univariate analysis, but in adjusted analyses the presence of a previous bacteraemia was a more important risk factor than antibiotic treatment itself. Manzoni *et al.*⁴² investigated risk factors for IFI in colonized infants. In the adjusted analysis, use of third-generation cephalosporins was not a risk factor for progression from colonization to invasive infection. We did not find it appropriate to pool the three RCTs reporting on associations between antibiotic exposure and IFI, due to the clinical and methodological diversity of studies and inclusion of moderately preterm infants, who nowadays are not viewed as having an increased risk of IFI.

Mortality

Twenty-one studies evaluated the risk of death after exposure to antibiotic treatment in the neonatal period (Table 3). There were 7 RCTs and 14 observational studies, of which 12 were retrospective. Kuppala *et al.*³⁰ reported on both duration and presence or absence of antibiotic treatment.

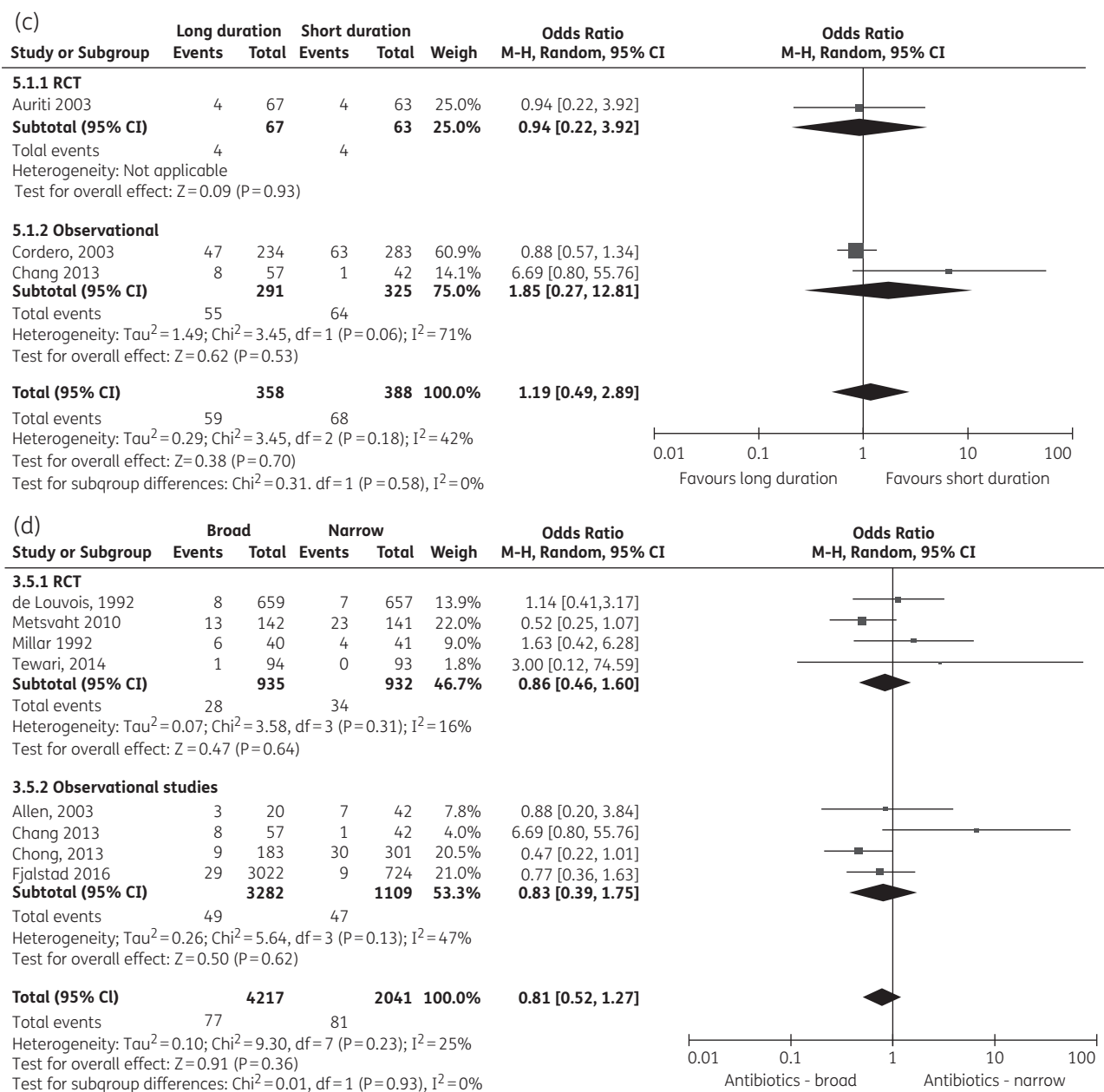


Figure 3. Continued

Death: antibiotics—yes/no

Three small studies were included,^{21,25,30} and they did not show any difference in mortality. Owing to clinical and methodological diversity between the studies we did not find it appropriate to pool data for meta-analysis.

Death: antibiotics—duration

Eleven studies were included.^{7,22,30–35,57,62,65} A large retrospective cohort study by Cotten *et al.*³³ reported that prolonged empirical antibiotic therapy (≥5 days) in patients without proven EOS was associated with an increased odds of death (OR 1.46, 95% CI

1.19–1.78) compared with shorter duration of antibiotic treatment, after adjusting for relevant confounders. In the same study,³³ each additional day of initial empirical antibiotic therapy increased the odds of death, a finding also reported by Abdel Ghany and Ali.³⁴ In a large Canadian cohort study of VLBW infants without infection-related morbidities in the first week of life, an increased risk of death was seen in infants (n = 4607) exposed to >3.5 days of antibiotics versus those (n = 3807) exposed to shorter duration, even after controlling for GA and severity of illness (OR 2.26, 95% CI 1.74–2.94).⁷ None of the other eight observational studies reported higher mortality (alone) with prolonged therapy. However, some of them were relatively small and therefore had limited power to

detect differences. Moreover, two large studies, by Cantey *et al.*⁶⁵ ($n = 2502$) and De Louvois *et al.*²⁶ ($n = 1316$), included predominantly term infants with a low risk of death. Kuppala *et al.*³⁰ reported that prolonged antibiotic therapy (≥ 5 days) increased the risk of the composite outcome LOS, NEC or death in VLBW infants. We pooled three studies^{22,31,62} with comparable groups but including <750 infants. In the pooled analysis (Figure 3c) the overall risk of death was not significantly increased after prolonged duration of antibiotic treatment, but again there was a moderate to substantial heterogeneity between study results.

Death: antibiotics—broad- versus narrow-spectrum regimen

Nine studies were included^{4,24,26–28,36,37,58,62} and the results were conflicting. A large observational study by Clark *et al.*³⁷ found a significant increase in the risk of death after use of ampicillin and cefotaxime versus ampicillin and gentamicin, despite the fact that patients receiving ampicillin and cefotaxime had a higher disease severity. We pooled the other eight studies (four RCTs and four observational studies) with comparable groups (Figure 3d). The two largest studies included predominantly term infants.^{4,26} Overall there were no significant differences in mortality between broad- and narrow-spectrum regimens and low to moderate heterogeneity between study results.

Discussion

Key findings

To our knowledge, this is the first systematic review examining serious adverse outcomes (NEC, IFI and death) in the neonatal period associated with antibiotic exposure early in life. One main finding is the clinical and methodological diversity of the 47 studies included and the lack of high-quality studies addressing these important adverse outcomes. Still, we believe there are three key findings of clinical relevance. First, five observational studies including >5000 infants showed that prolonged duration of antibiotic exposure in uninfected preterm infants is associated with an increased risk of developing NEC later in the neonatal period.^{30,33,34,55,56} Second, there is evidence from 10 observational studies, published since the year 2000, that previous exposure to third-generation cephalosporins or carbapenems is associated with an increased risk of developing IFI.^{32,40,47–50,60–63} Third, prolonged duration of empirical antibiotic therapy without any proven infection is also associated with increased all-cause mortality in preterm infants based on data from two very large observational trials^{7,33} and supported by three other observational studies.^{30,34,62} Few RCTs were identified in this systematic review. When restricting our meta-analysis to include only RCTs we found no significantly increased risk of NEC, IFI and/or death with our *a priori* comparisons of different categories of antibiotic exposure. However, most RCTs were old and even after pooling data from the RCTs the sample sizes were too small to detect clinically relevant differences in adverse outcomes between groups.

Biological plausibility

How could prolonged duration of antibiotic treatment in the neonatal period increase the risk of NEC and all-cause mortality?

Antibiotics perturb the early-life microbiota through their effects on the trajectory of microbial colonization, leading to a reduced pattern of microbial diversity and delayed commensal colonization, especially in preterm infants.^{57,66} Furthermore, the non-resilient early-life gut microbiota is more susceptible to antibiotic treatment, affecting the microbial composition to a greater extent, compared with the adult population.⁶⁷ This emphasizes the profound effects of early empirical antibiotic treatment in preterm infants. Although the pathogenesis of NEC is currently not well understood, bacterial colonization is thought to be a critical element in the development of the disease.^{55,68} Antibiotic therapy and its modulatory effects on the gut microbiota not only limit the diversity but may also lead to predominance of more pathogenic bacteria.⁶⁹ In neonates, shorter courses of antibiotic therapy (1–3 versus 5–7 days) are associated with a more rapid recovery from suppression of the gut microbiota.^{70,71} Taken together, these recent data may partly explain the increased risk of NEC in infants receiving prolonged empirical antibiotic therapy. Data from neonatal animal models also suggest an antibiotic-linked mechanism that places infants at higher risk of a humoral and cellular immune suppression.⁷¹ We found an increase in all-cause mortality after prolonged antibiotic therapy in preterm infants. This may be due to several reasons, including higher risk of NEC and increased risk of other later infections and/or immune-related diseases secondary to a certain degree of immune suppression.

Our review also suggests that exposure to third-generation cephalosporins or carbapenems increases the risk of IFIs, mainly *Candida* infections. Preterm infants are more prone to early colonization by *Candida* than term infants, due to an immature immune system and impaired skin and mucosal integrity.^{72,73} Furthermore, prolonged use of antibiotics may foster invasive *Candida* infections by suppressing normal flora and allowing *Candida* to occupy muco-epithelial niches that facilitate invasion and dissemination. *Candida albicans* may adapt to the intestinal environment through expression of genes that maintain their gut colonization following antibiotic-induced alterations in the bacterial microbiota.^{74,75} Cephalosporin use has been associated with intestinal colonization by *Candida* among neonates⁴⁴ and colonization is a risk factor for invasive *Candida* infections.^{41,76,77} Taken together, the results of the present review indicate that there are substantial data indicating that broad-spectrum antibiotics may pave the way for IFI.

Strengths and limitations

The strengths of our systematic review include our rigorous and sensitive search strategy following an *a priori* registered protocol. Additionally, we targeted an area of great clinical interest and concern: overuse of antibiotics in early life and adverse outcomes. The main limitations were that meta-analysis was only possible for a small subset of studies because most studies included were not randomized. Moreover, there was a substantial diversity among observational studies in terms of methodology, quality, sample size, type of antibiotic exposure and study design. Non-randomized studies are prone to biases that can be hard to identify and handle.¹⁶ In many of the large observational studies included in this systematic review the authors performed appropriate statistical adjustments for relevant clinical confounders, such as risk factors and illness severity. This reduces the risk of random findings in our review, but we cannot rule out residual confounding and

confounding by indication: sicker neonates receive more antibiotics but antibiotic exposure does not make them sicker.⁷ Overall, our intention was to collect as much evidence as possible related to the targeted adverse outcomes. This is why we elected to include non-randomized studies.^{15,16}

The evidence of a significant association between prolonged duration of antibiotic therapy and increased risk of NEC and/or death is therefore mainly supported by retrospective studies in preterm infants and we cannot conclude that there is a causal relationship. This also applies to the association between broad-spectrum antibiotics and increased risk of IFI. However, antibiotic exposure was identified before the outcomes and cohort studies potentially have a temporal framework to assess causality. We decided *a priori* to include studies with both term and preterm infants as we anticipated that some studies would include a mix of both, and we did not want to exclude these. However, very preterm infants have a much higher risk of developing the three severe adverse outcomes of interest compared with term infants. Some large studies included in this review failed to show an association between prolonged antibiotic therapy and adverse outcomes. These were often studies including predominantly 'low-risk' term infants, who only rarely develop NEC and IFI, and in general have a low mortality. The differences in study populations therefore need careful consideration when interpreting the results of our systematic review. We believe that, based on studies in our systematic review, it is possible to draw conclusions about the association between antibiotic exposure and adverse outcomes in preterm infants, whereas data on adverse outcomes in term infants are more limited and do not justify clear conclusions.

Finally, it was challenging in retrospect to label the antibiotic regimens of included studies as 'broad spectrum' or 'narrow spectrum'. In fact, most of those regimens labelled as narrow spectrum cover both Gram-positive and -negative bacteria, but their ecological profiles were more 'favourable' than regimens including third-generation cephalosporins and carbapenems. Lumping studies with different antibiotic combinations together still has clear limitations.

Implications and conclusion

This systematic review highlights the potential detrimental effect of unnecessary antibiotic treatment in the neonatal period, especially in preterm infants. Prolonged antibiotic therapy is associated with an increased risk of NEC and/or death in preterm infants. Measures should be taken to discontinue antibiotic treatment early if a clinically suspected infection is not confirmed.^{8,9} The results of our systematic review and other reviews^{7,8} also strongly advocate that the use of broad-spectrum antibiotics, particularly third-generation cephalosporins and carbapenems, should be restricted as they increase the risk of IFI, in addition to the increased risk of antibiotic resistance development.¹¹

Preventing infections, antibiotic stewardship and knowledge-based use of today's antibiotics are central principles to avoid overuse and adverse outcomes related to antibiotic exposure in the neonatal period, and to maintain safe and effective use of antibiotics in those who need treatment.

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Transparency declarations

None to declare.

Author contributions

E. E. reviewed all relevant titles, abstracts and full-text articles, assessed quality, extracted data and drafted the initial manuscript. J. W. F. reviewed all relevant titles, abstracts and full-text articles, extracted data, assessed quality and revised the manuscript. L. K. J. contributed to study design, carried out meta-analyses and revised the manuscript. J. N. vdA. contributed to study design, reviewed critically relevant full-text articles and revised the manuscript. C. K. conceptualized and designed the study, reviewed relevant abstracts and articles, assessed quality and revised the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work. E. E. and C. K. have full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Supplementary data

Tables S1–S3 and Appendixes A and B are available as Supplementary data at JAC online (<http://jac.oxfordjournals.org/>).

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